Frontend Quiz

JCWD-3302

1. What is the main role of HTML in web development ?
   a. **To define the structure and content of a web page**
   b. **To style the elements of a webpage**
   c. **To execute JavaScript code**
   d. **To store data in a database**

2. Which HTML element is used to create a hyperlink ?
   a. **<button>**
   b. **<link>**
   c. **<a>**
   d. **<img>**

3. What does the flex-grow property do in CSS Flexbox ?
   a. **Controls how much an element should grow to fill available space**
   b. **Defines the initial size of an element**
   c. **Specifies if an element can shrink**
   d. **Determines the main axis of the flex container**

4. What is `CSS Flexbox` and how is it used to layout elements on a web page?
   a. **CSS Flexbox is a method used for form validation in CSS.**
   b. **CSS Flexbox is a method for arranging the layout elements in one dimension (row or column) flexibly**
   c. **CSS Flexbox is a method for arranging background images within HTML elements.**
   d. **CSS Flexbox is a method for positioning element layout such as fixed, sticky, and absolute.**

5. What is the core concept of React that enables UI development using reusable pieces ?
   a. **MVC Pattern**
   b. **Component-Based Architecture**
   c. **Layout**
   d. **Object-Oriented Programming**

6. How do you define a functional component in React ?
   a. **class MyComponent extends React.Component {}**
   b. **const MyComponent = new React.Component();**
   c. **function MyComponent() { return <> </ > }**
   d. **React.createComponent('div', 'Hello')**

7. What is meant by `props drilling` in the context of React ?
   a. **Props drilling is a way to organize prop data in a structured and efficient manner**
   b. **Props drilling is a technique or method for passing props in React**
   c. **Props drilling is the process of passing props data through multiple layers of component, which can make the code inefficient.**
   d. **Props drilling is a method for passing props from child component to parent component in react, making the code efficient.**

8. What is the primary advantage of the Virtual DOM in React?
   a. **It reduces direct updates to the real DOM**
   b. **It increases the application's size**
   c. **It makes components larger**
   d. **It adds compatibility with HTML5**

9. How is useMemo used in React to optimize the performance of a component that fetches data from an external API ?
   a. **useMemo is used to efficiently fetch data from the API every time the component re-renders.**
   b. **useMemo is used to optimize performance by memoizing expensive operations or calculations that depend on values derived from API data.**
   c. **useMemo cannot be used to optimize data fetching from an API.**
   d. **useMemo is used to manage the rendering of React components.**

10. What is the purpose of the useState hook in React?
    a. **To manage routing**
    b. **To add state to functional components**
    c. **To handle component lifecycle**
    d. **To update the DOM manually**

11. Why is `useReducer` often used for state management instead of `useState` in more complex situations?
    a. **There is no correct answer.**
    b. **useReducer is simpler than useState in its implementation**
    c. **useReducer separates state updates from the component, making it easier to manage complex state.**
    d. **useState separates state updates from the component, making it easier to manage complex state.**

12. Which command is typically used to create a new React + TypeScript project using Vite?
    a. **npx create-react-app my-app --template typescript**
    b. **npm init react-typescript my-app**
    c. **npx vite-init react-ts my-app**
    d. **npm create vite@latest my-app -- --template react-ts**

13. Which hook is used to perform side effects in React?
    a. **useMemo**
    b. **useEffect**
    c. **useState**
    d. **useRef**

14. Which import is correct when using react-router-dom in vite reactjs project?
    a. **import { Switch, Route } from "react-router-dom";**
    b. **import { BrowserRouter, Routes, Route } from "react-router-dom";**
    c. **import { Router, Route } from "react-router";**
    d. **import { Navigation, Page } from "vite-router-dom";**

15. What is the correct TypeScript type for a component that receives children?
    a. **type Props = { children: JSX.Element };**
    b. **type Props = { children?: string };**
    c. **type Props = { children: React.ReactNode };**
    d. **type Props = { children: Element };**

16. What is the purpose of defining interfaces or types for props in React TypeScript?
    a. **To enable static type checking and avoid invalid props**
    b. **To automatically generate documentation**
    c. **To improve runtime performance**
    d. **To prevent React from re-rendering unnecessary components**

17. Which of the following is true about how state is updated in Zustand compared to Redux?
    a. **Redux uses mutation, while Zustand uses immutable updates**
    b. **Zustand uses reducers like Redux for all updates**
    c. **Redux requires pure functions (reducers), while Zustand allows direct state modification via set**
    d. Zustand uses dispatch to trigger actions like Redux

18. How do you access a React Context using useContext ?
    a. **const {value, setValue} = useContext(MyContext);**
    b. **const [state, dispatch] = useContext();**
    c. **const context = useContextProvider();**
    d. **useContext(MyContext, state => state.value);**

19. How do you define a Server Component in Next.js App Router ?
    a. **By using "use client" at the top of the file**
    b. **By omitting "use client" at the top of the file**
    c. **By exporting the component as ServerComponent**
    d. **By wrapping the component with useServer()**

20. Why should you avoid copying prop values into the component state?
    a. **Because props values cannot be stored by the state in that component**
    b. **Because it would create two instance with the same state, which could lead to state desynchronization.**
    c. **Because the mutation process on state does not allow such mutation processes**
    d. **Because it would lead to duplicate instances with different state, while state synchronization remains intact but it consume a lot of memory.**

21. What is meant by `Virtual DOM` in the context of react ?
    a. **A javascript library used to facilitate creating virtual animation interactions in react.**
    b. **A temporary representation of the actual DOM structure used by React to improve rendering performance.**
    c. **A type of component in react used to replace structure in the form of functional components**
    d. **A DOM element used to render components in React.**

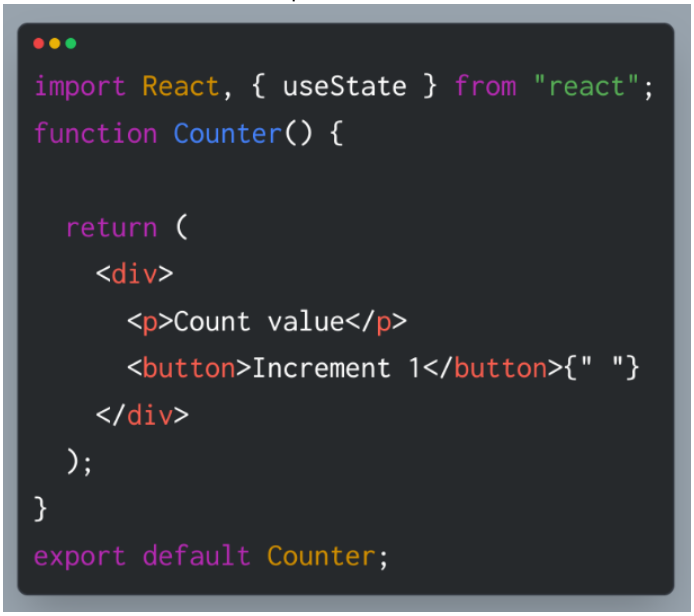
22. What is Server Side Rendering (SSR) in react, and what are its benefits?
    a. **SSR is a method for managing global state in react.**
    b. **SSR is a method for rendering views on the server side before sending them to the client, which allows the application to load faster and have better SEO.**
    c. **SSR is a method for rendering views only on the client side before sending them to the server, which allows the application to load faster and have better SEO**
    d. **SSR is a method for rendering only animated viewsin react before sending them to the server, which allow the applications to load faster and have better SEO**

23. Which of the following statements is correct regarding page navigation in a Next JS project?
    a. **Using Link from next/link is very good for SEO performance because page transition links can be detected by crawlers and have SPA behavior.**
    b. **There is no correct statement.**
    c. **Using Link from next/link is good for SEO performance because page transition links cannot be detected by crawlers.**
    d. **router.push is used for page navigation, which is very good for SEO performance because page transition links can be detected by crawlers and have SPA behavior**

24. Which method is used to fetch data on the server in a Server Component ?
    a. **fetch() inside useEffect**
    b. **getServerSideProps**
    c. **fetch() directly inside the component**
    d. **useSWR**

25. Which URL structure is more SEO-friendly ?
    a. **example.com/page?id=123&sort=asc**
    b. **example.com/?p=12345**
    c. **example.com/best-laptops-2024**
    d. **example.com/index.php?page=home**

26. Here is a functional component in react. How to complete this code with the correct way to manage state and display the state value when the button is pressed.

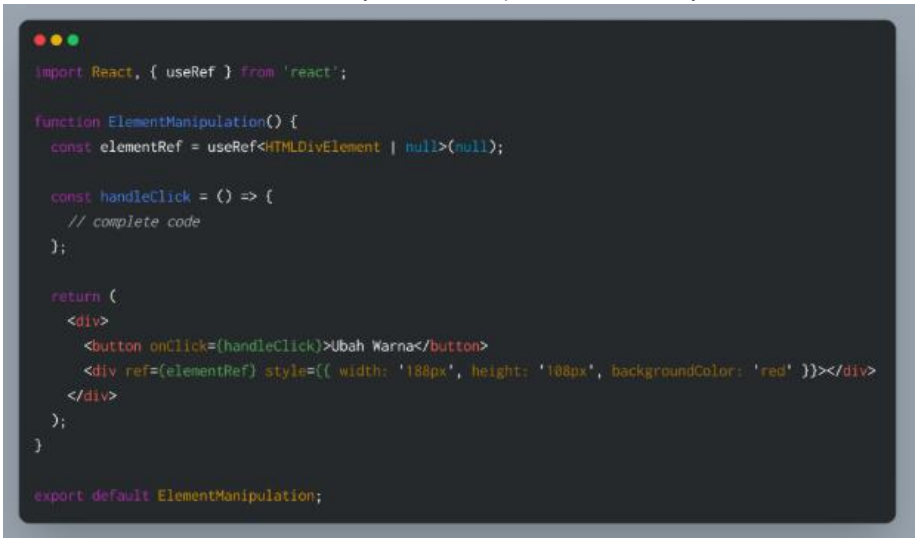


```
import React, { useState } from "react";
function Counter() {

  return (
    <div>
      <p>Count value</p>
      <button>Increment 1</button>{" "}
    </div>
  );
}
export default Counter;
```

a. **Use useState to initialize the count state and setCount, and use setCount to update the state when the button is pressed**
b. **State is not required in functional component**
c. **Use state to initialized the count state and setState, and use setState to update the state when the button is pressed.**
d. **Use this.state to initialize the count state and this.setState to update the state when the button is pressed**

27. You want to access DOM elements in your react component. How would you use the useRef hook to achieve this?



```
import React, { useRef } from 'react';

function ElementManipulation() {
  const elementRef = useRef<HTMLDivElement | null>(null);

  const handleClick = () => {
    // complete code
  };

  return (
    <div>
      <button onClick={handleClick}>Ubah Warna</button>
      <div ref={elementRef} style={{ width: '188px', height: '108px', backgroundColor: 'red' }}></div>
    </div>
  );
}

export default ElementManipulation;
```

a. **You can use elementRef.style.color = 'blue' in the handleClick functon to change the background color of the div element.**
b. **You can use elementRef.current.color = 'blue' in the handleClick function to change the background color of the div element.**
c. **You can use elementRef.style.backgroundColor = 'blue' in the handleClick function to change the background color of the div element**
d. **You can use elementRef.current.style.backgroundColor = 'blue' in the handleClick function to change the background color of the div element.**

28. You have a functional component in react that has several hooks

```jsx
import React, { useState, useEffect } from 'react';

function DataFetching() {
  const [data, setData] = useState<any[]>([]);
  const [loading, setLoading] = useState<boolean>(true);

  useEffect(() => {
    // complete code
  }, []);

  return (
    <div>
      <h1>Data yang diambil:</h1> {/* Corrected heading tag */}
      {loading ? (
        <p>Loading...</p>
      ) : (
        <ul>
          {data.map((item) => (
            <li key={item.id}>{item.title}</li>
          ))}
        </ul>
      )}
    </div>
  );
}

export default DataFetching;
```

Complete the following code:

a. **You can directly use data and loading to display data in the UI without needing to use useEffect**
b. **In useEffect, you will send a HTTP request using fetch to fetch data from the server and populate data with the response. When the data is received, you will change the loading value to false**
c. **You should add and create a new state inside useEffect to manage data and loading because you cannot use useState outsidethe return. After that, make an HTTP request directly inside useEffect.**
d. **You should initialize data and loading with fake data from the start, and then update them when the actual data arrives.**

29. You have a functional component in React that has several hooks

```tsx
import React, { useState, useEffect } from 'react';

function DataFetching() {
  const [data, setData] = useState<any[]>([]);
  const [loading, setLoading] = useState<boolean>(true);

  useEffect(() => {
    // complete code
  }, []);

  return (
    <div>
      <h1>Data yang diambil:</h1> {/* Corrected heading tag */}
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        </ul>
      )}
    </div>
  );
}

export default DataFetching;
```

Complete the following code:

a. **You can directly use data and loading to display data in the UI without needing to use useEffect/**
b. **In useEffect, you will send a HTTP request using fetch to fetch data from the server and populate data with the response. When the data is received, you will change the loading value to false/**
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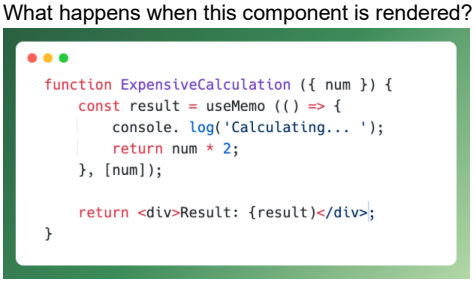
30. What happens when this component is rendered?

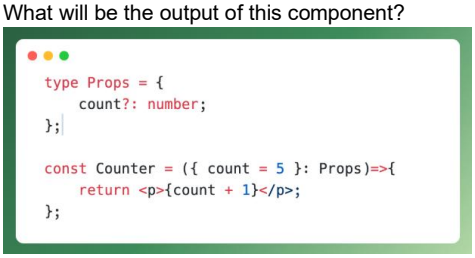


```
function ExpensiveCalculation ({ num }) {
    const result = useMemo (() => {
        console. log('Calculating... ');
        return num * 2;
    }, [num]);

    return <div>Result: {result)</div>;
}
```

a. **Logs 'Calculating...' on every render**
b. **Logs nothing**
c. **Logs 'Calculating...' only when num changes**
d. **Throws an error**

31. What will be the output of this component?



```
type Props = {
    count?: number;
};

const Counter = ({ count = 5 }: Props)=>{
    return <p>{count + 1}</p>;
};
```

a. **It will display 5**
b. **It throws a TypeScript error because count is optional**
c. **It will display 6**
d. **It displays undefined1**

32. How do you properly define global state and actions using Zustand?
a. **By using `createSlice` to organize reducers and actions.**
b. **By using the `create` function from Zustand to define a store that returns an object with state and actions.**
c. **By defining a root reducer and using `combineReducers` to manage state structure.**
d. **By using `useReducer` inside React Context to manage state globally.**

33. Take a look at the following code! The snippet below is from a Next.js version 14 project that uses the app router.

```tsx
import { useState } from "react";

export default function Home() {
  const [data, setData] = useState<number>(0);

  return (
    <div>
      <button onClick={() => setData(data + 1)}>+</button>
      <div>{data}</div>
    </div>
  );
}
```

What is wrong with the code ?

a. **You need to define the page using class component .**
b. **You need to define the page as client side using "use client" .**
c. **The use of useState must be accompanied by useEffect in that component.**
d. **The Function in the onClick must be wrapped in seperate function and initialized above before calling it.**